Evolutionary Computation

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Introduction to EC

- General term for several computational techniques inspired by biological evolution

- Evolutionary Computation is the field of study devoted to the design, development, and analysis of problem solvers based on natural selection (simulated evolution).

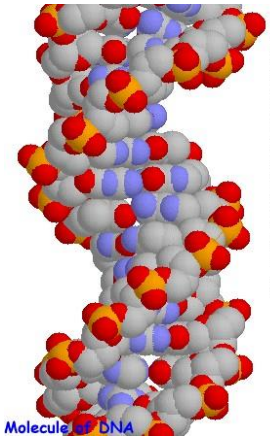
Introduction to EC

- General EC metaphor:

  a) A population of individuals exists in an environment with limited resources
  b) *Competition* for those resources causes selection of those *fitter* individuals that are better adapted to the environment
  c) These individuals act as seeds for the generation of new individuals.
  d) The new individuals have their fitness evaluated and compete (possibly also with parents) for survival.
  e) Over time *natural selection* causes a rise in the fitness of the population

Introduction to EC

- EC mostly involve meta-heuristic optimization algorithms such as:
  - Evolutionary algorithms
    - comprising genetic algorithms, evolutionary programming, etc)
  - Swarm intelligence
    - comprising ant colony optimization, particle swarm optimization, etc)





GA & PSO will be explained in details in the following chapter

General Steps in EC

1. In an Evolutionary Computation, a population of candidate solutions (CSs) is randomly generated.

2. Each of the CSs is evaluated and assigned a fitness based on a user specified evaluation function. The evaluation function is used to determine the ‘goodness’ of a CS.

3. A number of individuals are then selected to be parents based on their fitness. The *Select_Parents* method must be one that balances the urge for selecting the best performing CSs with the need for population diversity.

General Steps in EC

4. The selected parents are then allowed to create a set of offspring which are evaluated and assigned a fitness using the same evaluation function defined by the user.

5. A decision must be made as to which individuals of the current population and the offspring population should be allowed to survive. Typically, in EC , this is done to guarantee that the population size remains constant. [The study of ECs with dynamic population is an interesting project]

6. Once a decision is made the survivors comprise the next generation (Pop(t+1).

General Steps in EC

7. This process of selecting parents based on their fitness, allowing them to create offspring, and replacing weaker members of the population is repeated for a user specified number of cycles.

8. Stopping conditions for evolutionary search could be:

    a) The discovery of an optimal or near optimal solution
    b) Convergence on a single solution or set of similar solutions,
    c) When the EC detects the problem has no feasible solution,
    d) After a user-specified threshold has been reached, or
    e) After a maximum number of cycles.

EC Pseudo-code

```
Procedure EC{
  t = 0;
  Initialize Pop(t);
  Evaluate Pop(t);
  While (Not Done)
  {
    Parents(t) = Select_Parents(Pop(t));
    Offspring(t) = Procreate(Parents(t));
    Evaluate(Offspring(t));
    Pop(t+1)= Replace(Pop(t),Offspring(t));
    t = t + 1;
  }
```

EC Domain of Applications

1. Numerical, Combinatorial Optimization
2. System Modeling and Identification
3. Planning and Control
4. Engineering Design
5. Data Mining
6. Machine Learning

EC Advantages

1. Widely applicable
2. Low development & application costs
3. Easy to incorporate other methods
4. Solutions are interpretable (unlike NN)
5. Can be run interactively, accommodate user proposed solutions
6. Provide many alternative solutions

EC Disadvantages

1. No guarantee for optimal solution within finite time
2. Weak theoretical basis
3. May need parameter tuning
4. Often computationally expensive, i.e. slow